

TECHNICAL MEMORANDUM

Doc Ock: A Tendon-Driven Logarithmic-Spiral Gripper with IMU Teleoperation

After the SpiRobs design of Wang, Freris, and Wei [1]

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ABSTRACT. Doc Ock is a gripper shaped as a logarithmic spiral, modeled on an octopus arm and derived from the SpiRobs work [1]. Three independent cables give three degrees of freedom; differential cable tension curls the spiral around an object. An operator drives it by moving a handheld IMU. Luke Salazar led the CAD; the author owned the electronics and firmware. The project is paused with no surviving hardware, and its limitations are stated in Section 5.

1 Why a Spiral

A parallel-jaw gripper holds an object at two contact patches and depends on the object presenting two roughly parallel faces. A logarithmic spiral conforms: as the arm curls it wraps the object and contacts it along its length, so irregular and non-uniform objects that defeat parallel jaws can be held. Wang, Freris, and Wei showed the same geometry grasping across more than two orders of magnitude in object size [1]. Doc Ock reproduces that geometry at desk scale with printed links and cable actuation.

2 Mechanical Design

Every link was custom-designed in SolidWorks. Links are joined by TPU interlink plugs, making the arm a multi-material assembly: rigid printed links for the structure and compliant plugs at the joints so the spiral can flex under cable tension and return.

3 Actuation

Three cables run the length of the arm, each driven by its own NEMA17 stepper through a TMC2209 driver under UART control. Pulling one cable curls the arm toward it; pulling them differentially sets the curl's direction and tightness. The three cables together give the arm three degrees of freedom.

4 Control

The arm is teleoperated from an MPU6050 IMU held by the operator. An ESP32 reads the IMU and maps its motion to the three motors: translation along X drives motor 1, Y drives motor 2, and Z drives motor 3. The operator moves the IMU and the gripper follows.

Table 1. Teleoperation mapping and system summary.

IMU axis	Motor	Drive
X translation	motor 1, NEMA17	TMC2209, UART
Y translation	motor 2, NEMA17	TMC2209, UART
Z translation	motor 3, NEMA17	TMC2209, UART
Controller ESP32; operator input MPU6050 IMU; links SolidWorks, FDM; joints TPU interlink plugs		

5 Known Limitations

- No closed-loop tendon tension feedback. Cable tension is set open-loop by stepper position.
- Reachable workspace is unquantified.
- Failure modes under load are uncharacterized.
- Grip force, object size range, and link count were not recorded.

6 Status and Planned Work

The project is fully paused and no hardware survives. The planned next step was a design optimization study covering reachable workspace, manufacturability, and points of failure.

7 Role and Attribution

Co-contribution with Luke Salazar. Luke led the CAD. The author owned the electronics and firmware, and learned CAD technique from Luke over the course of the build.

References

1. Z. Wang, N. M. Freris, and X. Wei, "SpiRobs: Logarithmic spiral-shaped robots for versatile grasping across scales," *Device*, Cell Press, 2024. arXiv:2303.09861.